MODULE – III

DATA VISUALIZATION

Course Name: **DATA EXPLORATION AND PREPARATION**
Course Code : [23CSE617]
Total Hours : **09**

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Syllabus

Principles of Data Visualization: Types of Data Visualization, Choosing right chart for data, Best practices for Effective Visualization. Data Visualization Libraries: Seaborn Library for Statistical Analysis, Matplotlib Library, Plotly Express Library.

Principles of Data Visualization

- Data visualization is the graphical representation of information and data in a pictorial or graphical format (Visualization of Data could be: charts, graphs, and maps)
- It simplifies the complex quantitative information
- It helps to analyse and explore big data easily.
- It identifies the area that need attention or improvement
- It identifies the relationship between datapoints and variables
- Explores new patterns and revels hidden pattern in the data

Considerations - Three major consideration for data visualization are:

i. **Clarity -** Ensure the dataset is complete and relevant. This enables the data scientist to use the patterns yield from the data in the relevant places.

ii. **Accuracy -** Ensure using appropriate graphical representation to convey the right message

iii. **Efficiency -** Use efficient visualization technique which highlights all the data points

1. Types of Data Visualization

Summary

- **Temporal** – Visualizations that display data over time, such as trends, changes, or patterns across days, months, or years.
- **Hierarchical** – Visualizations that represent data organized into levels or parent-child relationships, showing how smaller parts relate to a larger whole.
- **Network** – Visualizations that illustrate relationships or connections between entities using nodes and links.
- **Multi-dimensional** – Visualizations that present data with multiple variables simultaneously to reveal patterns, correlations, or comparisons.
- **Geospatial** – Visualizations that map data to geographic locations to show spatial patterns and relationships.
- **Miscellaneous** – Visualizations that do not fit into a specific category but are used to present unique, specialized, or combined data representations.

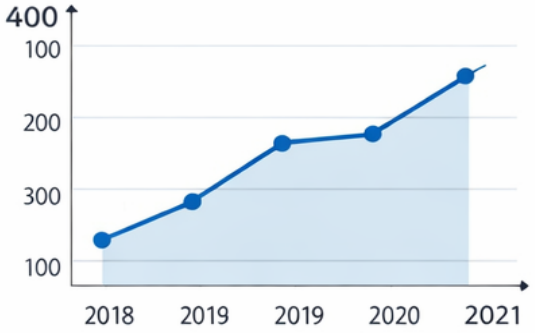
1. Types of Data Visualization

Temporal Visualizations (Time-Based Data)

- Line Graph – Shows trends and changes in data over time.
- Timeline – Displays events in chronological order.
- Kagi Chart – Identifies trend direction and reversals (used in financial analysis).
- Stacked Area Chart – Shows cumulative totals over time with category contribution.
- Gantt Chart – Represents project tasks and schedules across time.

Line Graph

Trends Over Time



Sales Growth Over Time

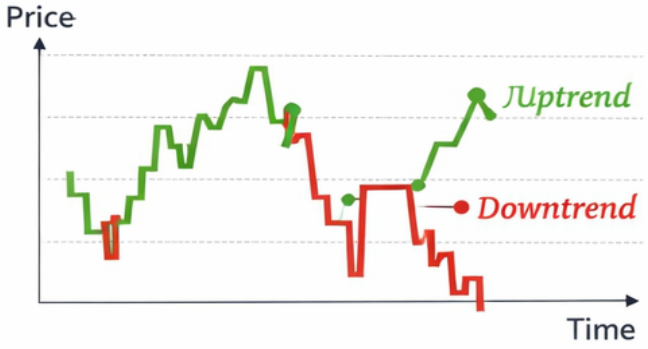
Timeline

Project Milestones

Project Timeline

Kagi Chart

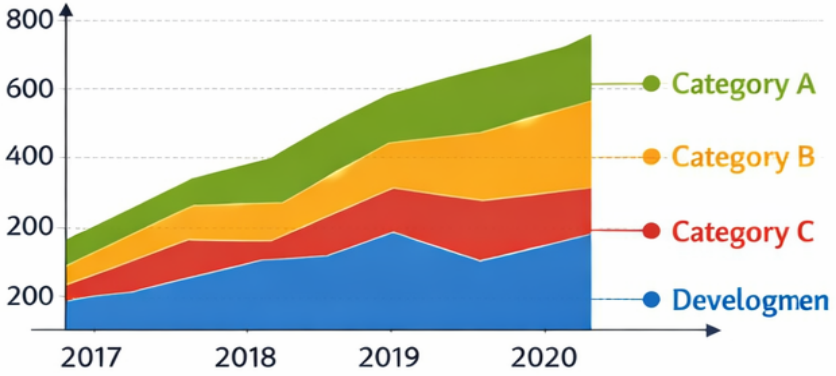
Price Trend Reversals



Stock Price Changes

Stacked Area Chart

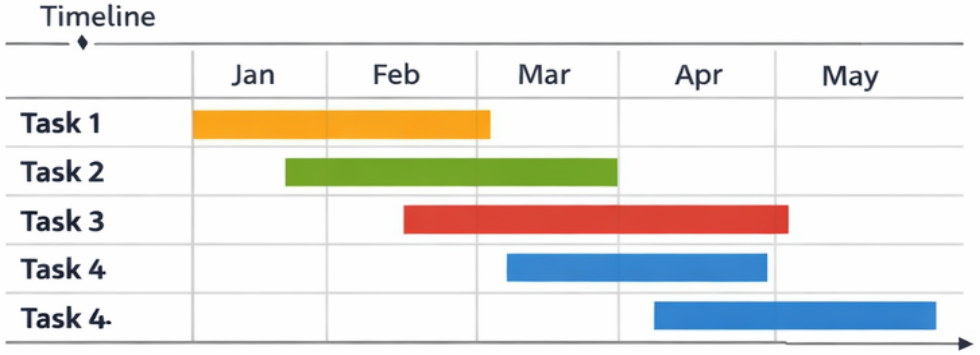
Cumulative Data



Revenue Breakdown

Gantt Chart

Project Schedule



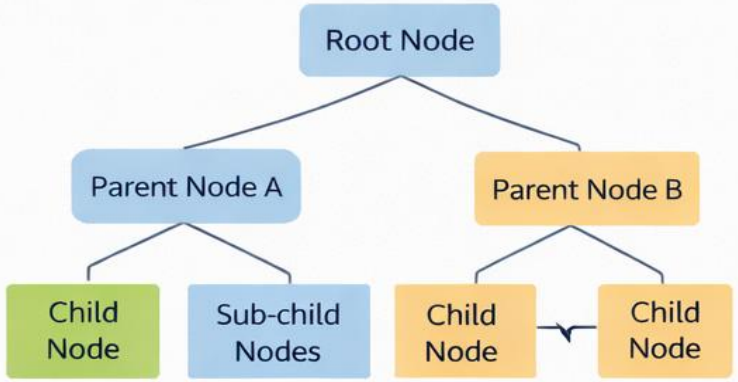
Display: Intercomections batween goops in in grid-format

Hierarchical Visualizations (Structured / Parent–Child Data)

- Tree Diagram – Displays hierarchical parent-child relationships.
- Tree Map – Shows hierarchical proportions using nested rectangles.
- Sunburst (Ring) Chart – Represents hierarchical levels in concentric circles.
- Circle Packing – Visualizes hierarchy using nested circles.
- Dendrogram – Displays hierarchical clustering relationships.

Tree Diagram

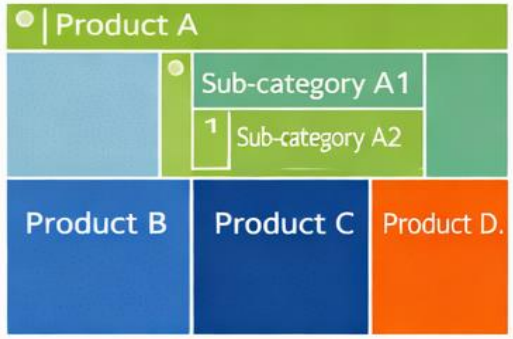
Hierarchical Relationships



Hierarchical Relationships

Tree Map

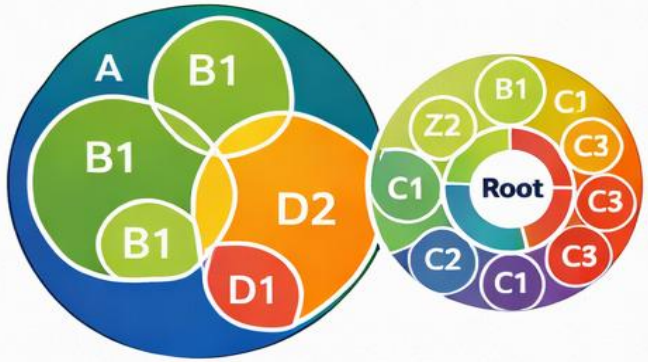
Hierarchical Proportions



Hierarchical Proportions

Sunburst (Ring) Chart

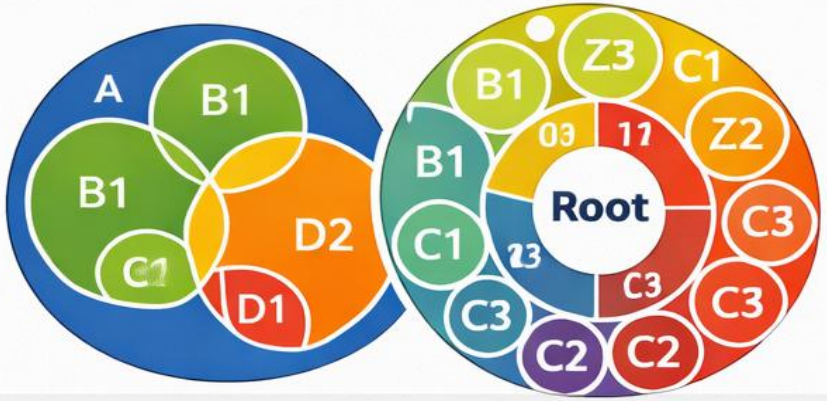
Hierarchical Levels in Concentric Circles



Hierarchical Levels in Concentric Circles

Circle Packing

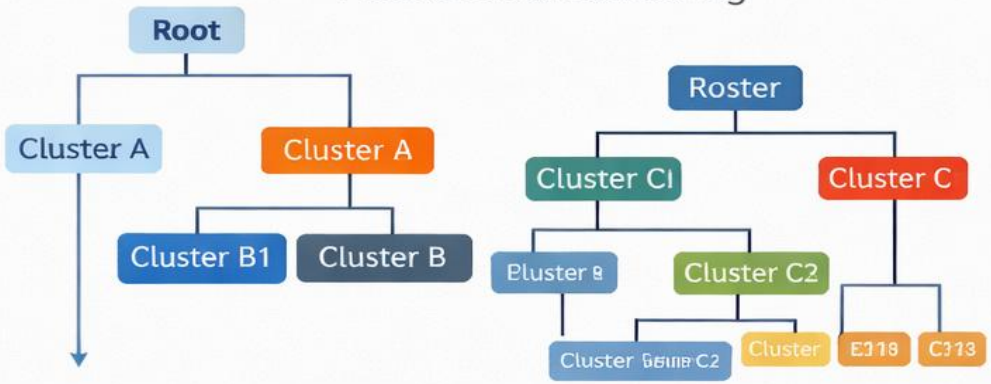
Herarchical Hierarchy



Nested Circles Hierarchy

Dendrogram

Hierarchical Clustering



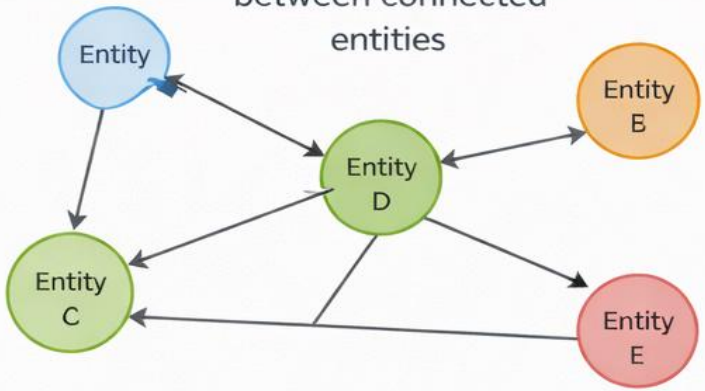
Hierarchical Clustering

Network Visualizations (Relationships & Connections)

- Node-Link Diagram – Shows relationships between connected entities.
- Alluvial Diagram – Displays flow and category changes across stages.
- Flow Map – Shows movement between locations or entities.
- Matrix Chart – Represents relationships between two sets in grid format.
- Chord Diagram – Displays interconnections between groups in circular form.

Node-Link Diagram

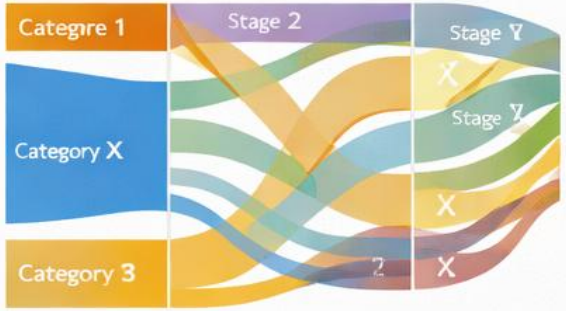
Shows relationships between connected entities



Shows relationships between connected entities

Alluvial Diagram

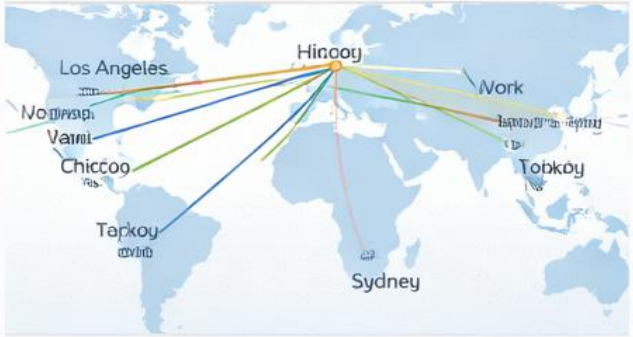
Displays flow and category changes across stages



Displays flow and category changes across stages

Flow Map

Shows movement between locations or entities



Shows movement between locations or entities

Matrix Chart

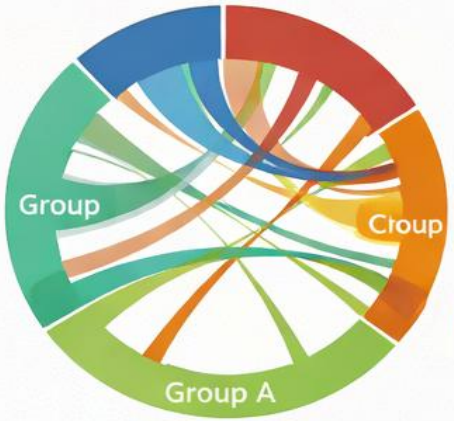
Represent sutarnships

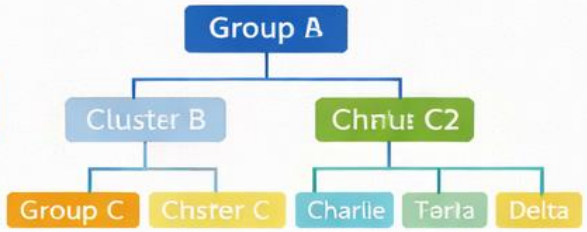
	Alpha	Bravo	Charlie	Delta	Delta
Alpha	●	●	●	●	●
Bravo	●	●		●	●
Charlie		●	●	●	●

Represents relationships between two sets in grid format

Chord Diagram

Displays interconnections between groups in circular form



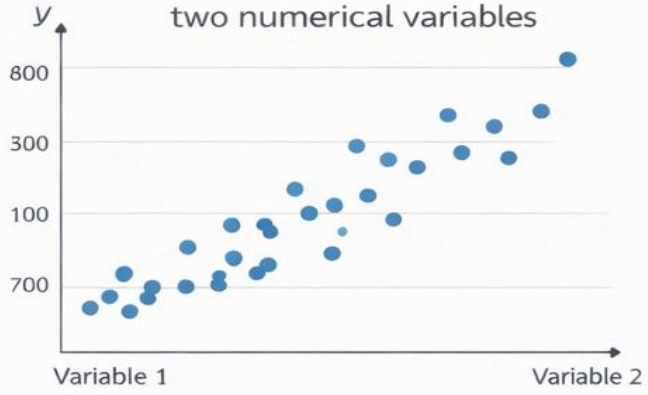


Multi-Dimensional Visualizations (Multiple Variables Analysis)

- Scatter Plot – Shows relationship between two numerical variables.
- Bubble Chart – Adds third variable using bubble size.
- Parallel Coordinate Plot – Visualizes multiple variables simultaneously.
- Box Plot – Shows distribution, median, quartiles, and outliers.
- Radar Chart – Compares multiple quantitative variables in radial format.
- Stacked Bar Graph – Compares multiple subcategories within categories.

Scatter Plot

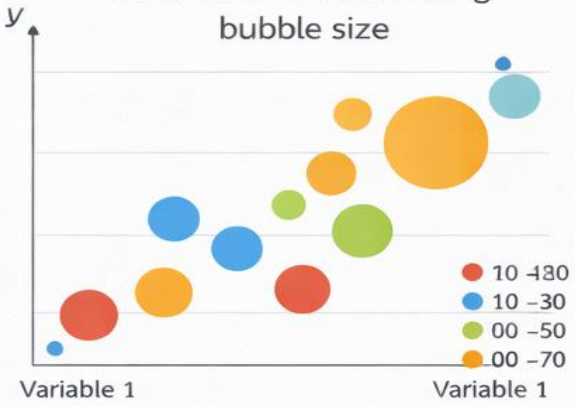
Shows relationship between two numerical variables



Shows relationship between connectrel entites

Bubble Chart

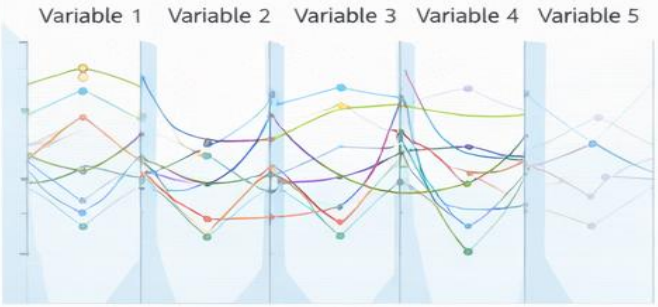
Adds third variable using bubble size



Adds third variable using bubble size

Parallel Coordinate Plot

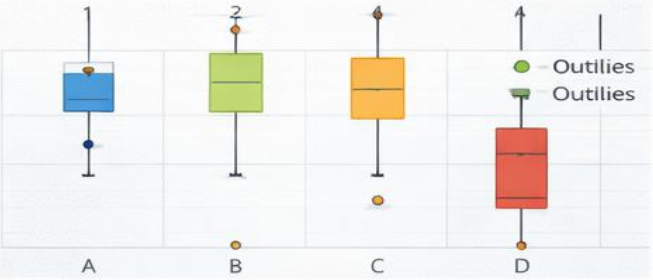
Visualiizes multiple variables simutianetssily



Visualizes multiple variables simuliametusly

Box Plot

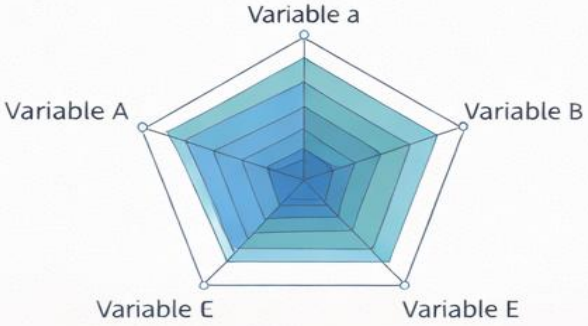
Shows distribution, median, quartiles, and outliers



Shows distribution, median, quartiles, and outliers

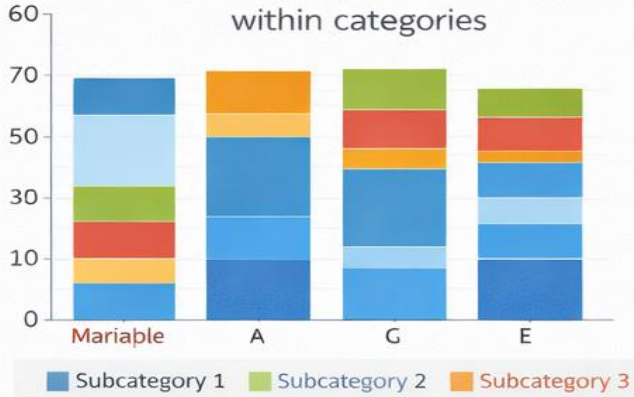
Radar Chart

Compares multiple quantiative variables in radial format



Stacked Bar Graph

Compares multiple subcategories within categories

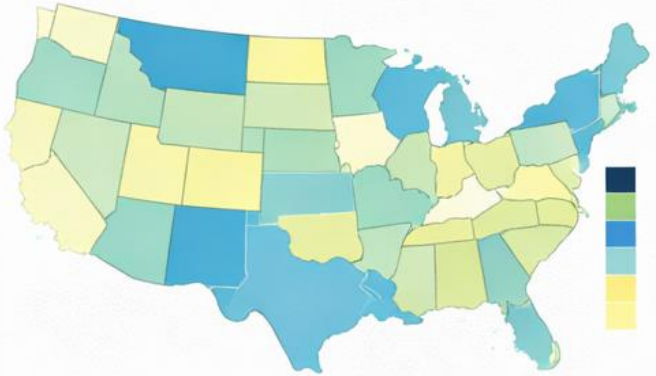


Geospatial Visualizations (Location-Based Data)

- Choropleth Map – Uses color shading to represent regional values.
- Cartogram – Distorts geographic size based on data magnitude.
- Heat Map (Geographic) – Shows density or intensity across locations.
- Dot Density Map – Displays concentration using dots in geographic areas.
- Proportional Symbol Map – Uses symbol size to represent data magnitude at locations.

Choropleth Map

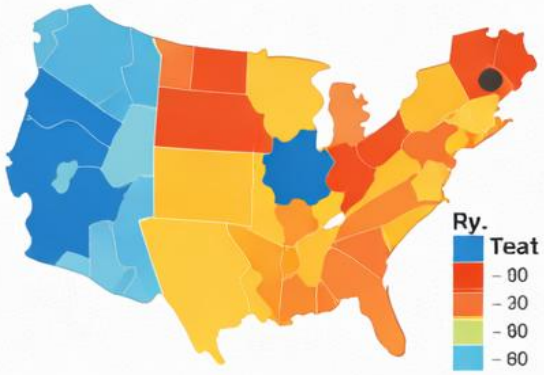
Uses color shading to represent regional values



Uses color shading to represent regional values

Cartogram

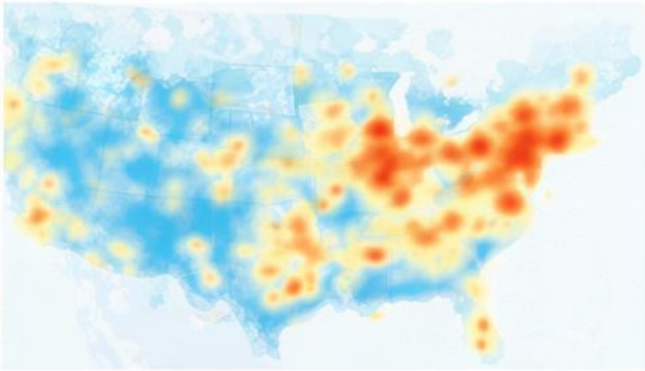
Distorts geographic size based on data magnitude



Distorts geographic size based on data magnitude

Heat Map (Geographic)

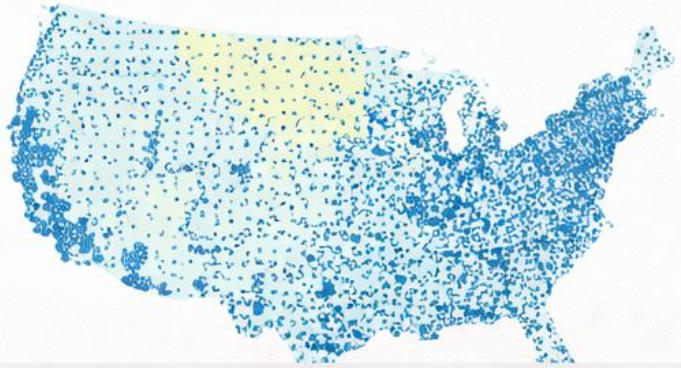
Shows density or intensity across locations



Shows density or intensity across locations

Dot Density Map

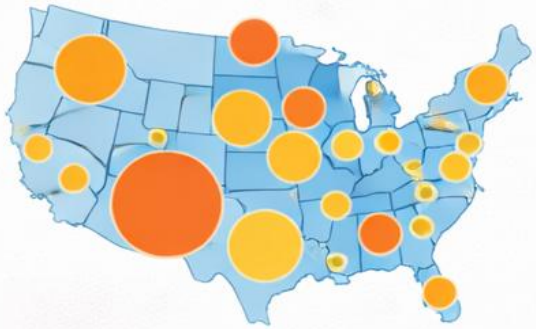
Displays concentration using dots in geographic areas

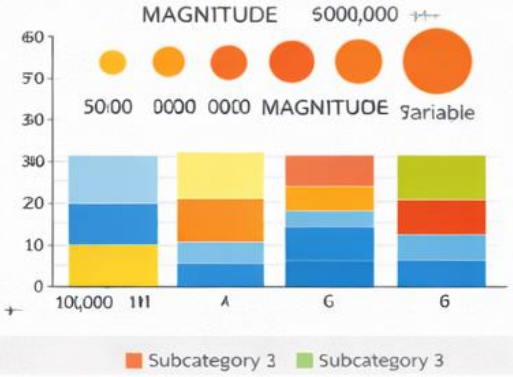


Displays concentration using dots in geographic areas

Proportional Symbol Map

Uses symbol size to represent data magnitude at locations





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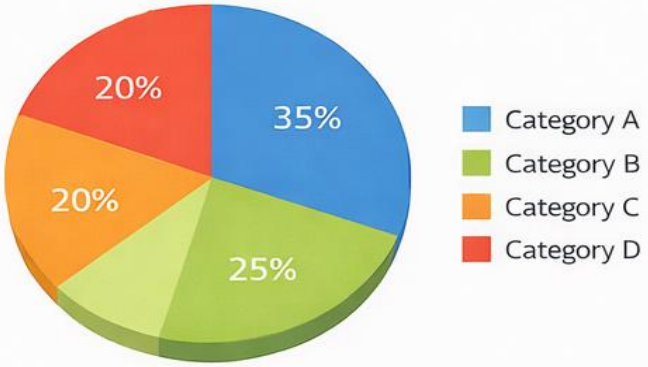
Miscellaneous / General Purpose Visualizations

- Pie Chart – Shows percentage distribution of a whole.
- Polar Area Diagram – Displays cyclic or categorical data in radial form.
- Word Cloud – Represents text frequency visually.
- Funnel Chart – Shows progressive reduction across process stages.
- Histogram – Displays frequency distribution of continuous data.
- Waterfall Chart – Shows cumulative effect of sequential positive and negative values.

Pie Chart

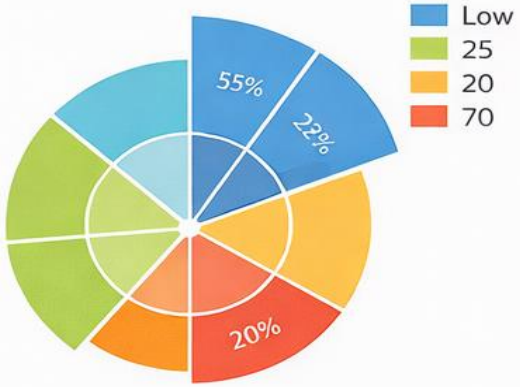
Shows percentage distribution of a whole



Shows percentage distribution of a whole

Polar Area Diagram

Displays cyclic or categorical data in radial form



Displays cyclic or categorical data in radial form

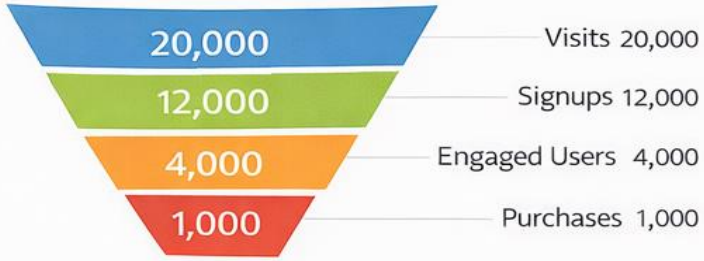
Word Cloud

Represents text frequency visually

Represents text frequency visually

Funnel Chart

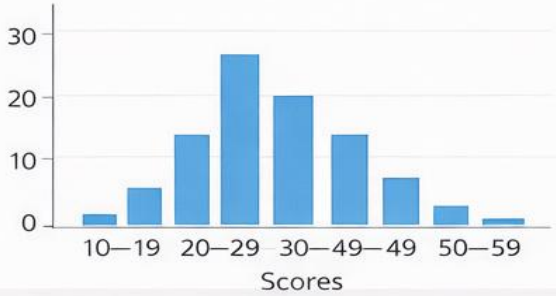
Shows progressive reduction across process stages



Displays concenration reduction across process stages

Histogram

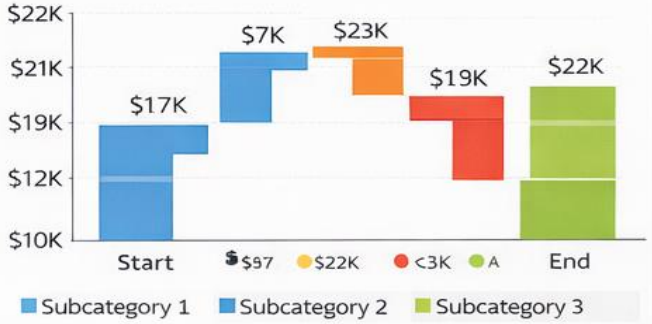
Displays frequency distribution of continuous data



Displays frequency distribution of continuous data

Waterfall Chart

Shows cumulative effect of sequential positive and negative values



2. Choosing right chart for data

https://flourish.studio/blog/choosing-the-right-visualisation/

- Selecting the appropriate chart type is crucial for crafting a clear data visualization.
- Making an incorrect choice can lead to misinterpretations, confusion, and hinder the audience's understanding of your intended message.
- With numerous chart types available, the selection process may initially seem daunting, but it becomes much more manageable once you understand the guiding principles.
- When deciding which chart to use, you should always start with the most important question: *"What are you trying to communicate with your data? What is the story you want to tell?"*
- Do you want to show change over time? A list of rankings? Correlation between two variables? Once you know that, the options are clearer.

Choosing right chart for data

- This topic presents nine different types of information that you can display with a visualization. This isn't a comprehensive list, and there are bound to be exceptions to these categories.
- With experience you will be able to more quickly assess what chart type you want to create. Because Tableau is flexible, we encourage you to think outside the box.
- However, before you think outside the box, it's helpful to start with some common chart types.

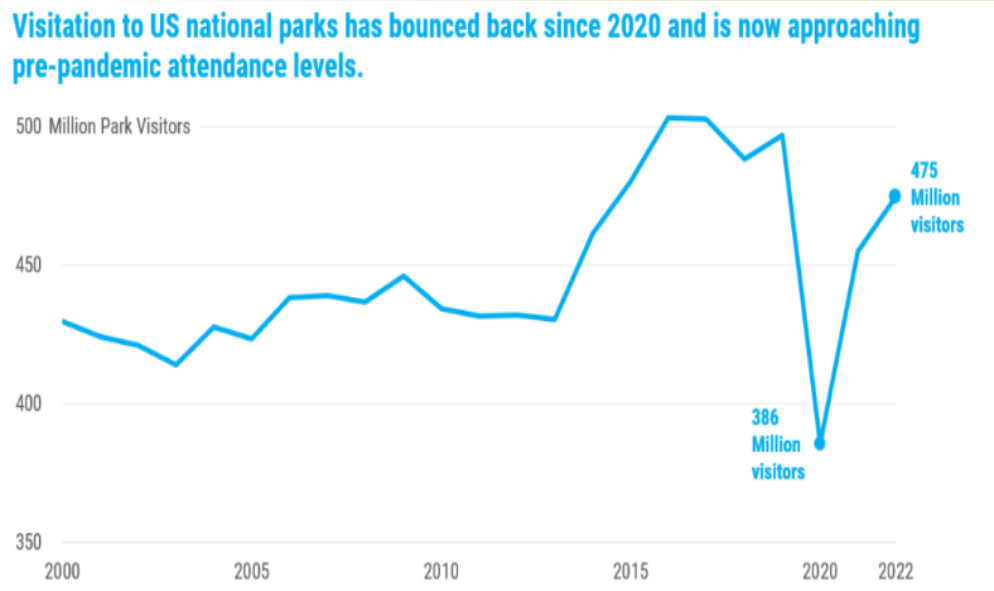
Choosing right chart for data

1. Change over time

- Showing a change over time for a measure is one of the fundamental categories of visualizations. There are many options for exploring change over time, including line charts, slope charts, and highlight tables.
- To show change over time, you need to know the value you expect to change, and how to work with Date fields in Tableau.

What kind of question does this chart answer?

- How has this measure changed in the past year?
- When did this measure change?
- How quickly has this measure changed?



Choosing right chart for data

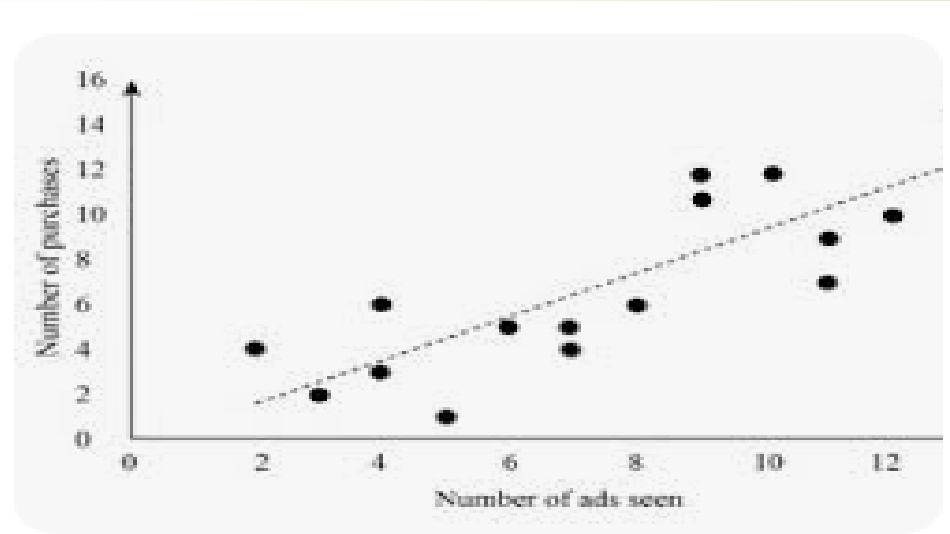
2. Correlation

- Sometimes you have two variables and are looking for the relationship between them. For example, you may be looking for the relationship between classroom size and school graduation rate, or how much lung capacity relates to endurance. (But remember, correlation does not always equal causation.)

- Correlation can be shown with scatter plots or highlight tables, and you can use Tableau's analytics objects(Link opens in a new window) to show the strength of the correlation.

What kind of question does this chart answer?

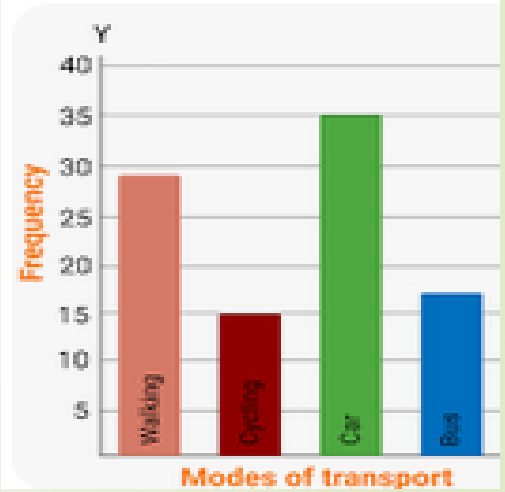
- Are these two measures related? How strongly?
- Are some measures more related than others?
- How strongly related are these measures?



Choosing right chart for data

3. Magnitude

- Magnitude shows the relative size or value of two or more discrete items. If you are comparing sales for different regions, you are looking at magnitude. Magnitude charts include bar charts, packed bubble charts, and line charts.



What kind of question does this chart answer?

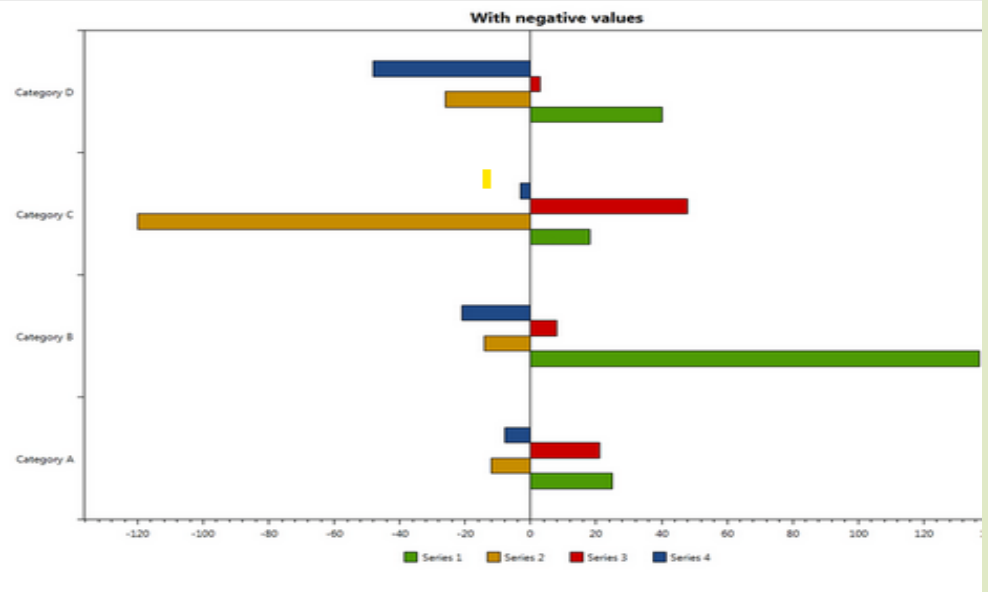
- Which of these dimension members has the highest measure?
- Are there any exceptional dimensions?
- How large of a gap is there between the lowest and highest measure between these dimensions?

Choosing right chart for data

4. Deviation

- Deviation charts show how far a value varies from some baseline, such as the average or median. If you wanted to know which items had unusually high or low profit margins, you would use a deviation chart.

- You can use bullet charts, bar charts, and combination charts to show deviation. You can also find the statistical significance of the deviation using a Z-score.



What kind of question does this chart answer?

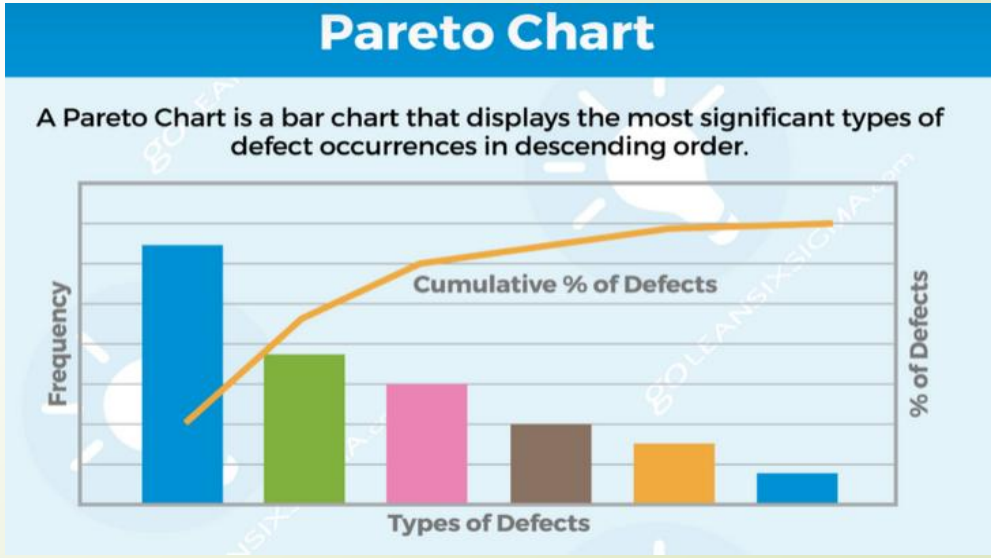
- How far from the norm does this measure stray?
- How important are the deviations in this measure?
- Is there a pattern to the deviations?

Choosing right chart for data

5. Distribution

- When you are trying to find the frequency of events within a population, you are looking at the distribution. If you are showing the number of respondents to a survey by age, or the frequency of incoming calls by day, a distribution chart might be the best choice.

- Distribution charts include histograms, population pyramids, Pareto charts, and box plots.



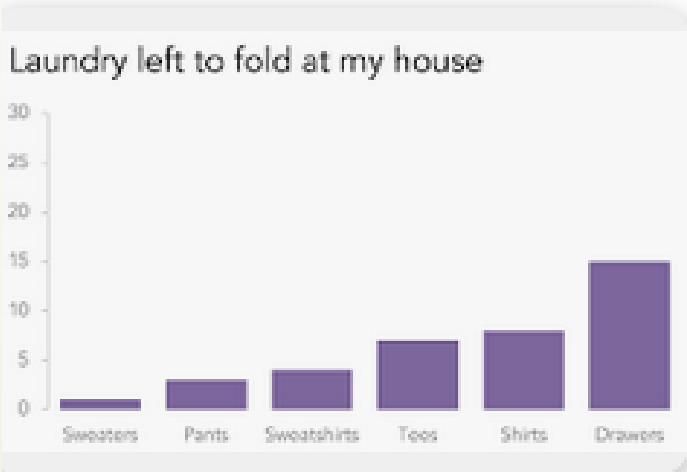
What kind of question does this chart answer?

- Are events clustered around a certain probability?
- Which population group buys the most items?
- When are the busiest times in our work day?

Choosing right chart for data

6. Ranking

- Sometimes you not only want to depict the magnitude of some value, but also the relative ranking of all the members of your dimension. Showing the top ten sales people or demonstrating the under-performing states use a ranking chart.
- Ranking charts are usually bar charts that integrate rank calculations, top n sets, or key progress indicators.



What kind of question does this chart answer?

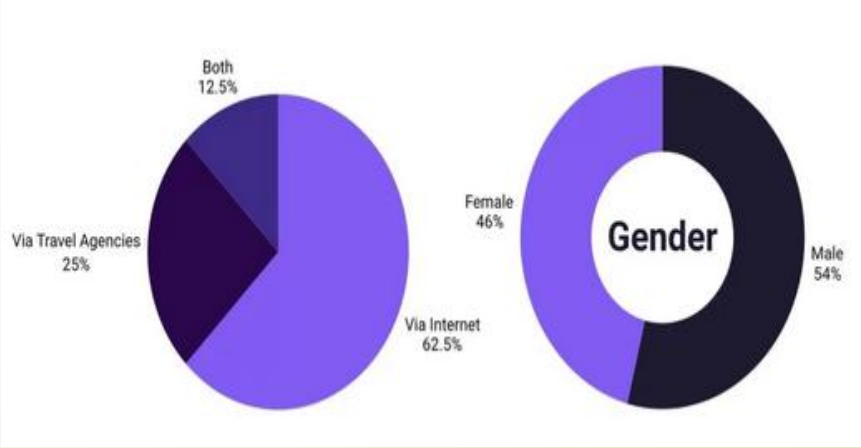
- How many people are under-performing in the company?
- How much revenue is generated by our top ten customers?
- What is the value of our ten lowest revenue properties?

Choosing right chart for data

7. Part-to-Whole

- Part-to-Whole charts show how much of a whole an individual part takes up. For example, if you are showing how much each region contributes to overall sales, or how expensive each different shipping mode is for an individual product, you would use a part to whole chart

- Part-to-Whole charts can be pie charts, area charts, stacked bar charts, or treemaps.



What kind of question does this chart answer?

- How much does this value contribute to the total?
- How does the distribution of costs change each year?
- Do different items contribute different amounts to sales by region?

Choosing right chart for data

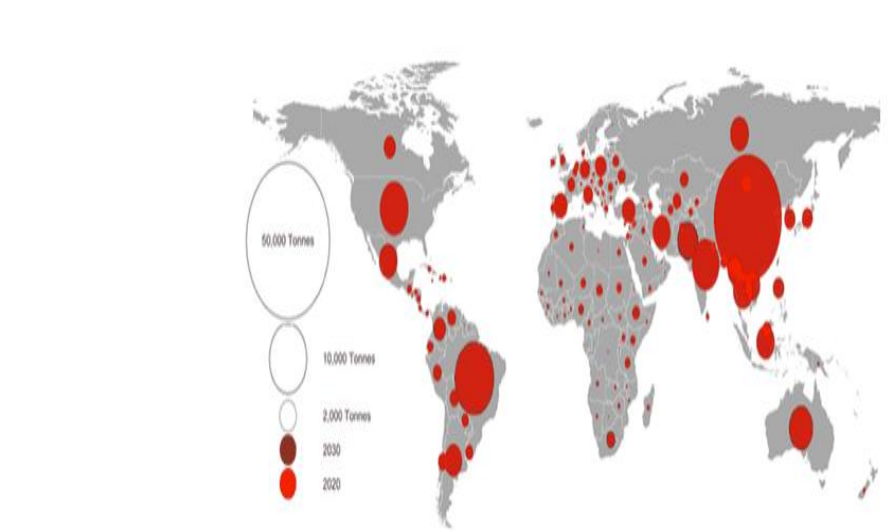
8. Spatial

- Spatial charts can precise locations and geographical patterns in your data. Showing the airport terminals with the most foot traffic or a map of all sales across the country are examples of spatial maps.

- Spatial maps include filled maps, point distribution maps, symbol maps, and density maps.

What kind of question does this chart answer?

- Which city has the highest sales?
- How far from distribution centers are our customers?
- How many people arrive at which gate?



Graduated symbol map showing antimicrobial use in 2020 (red circles) and projected increase by 2030 (dark red rings). Source: PLOS Glo

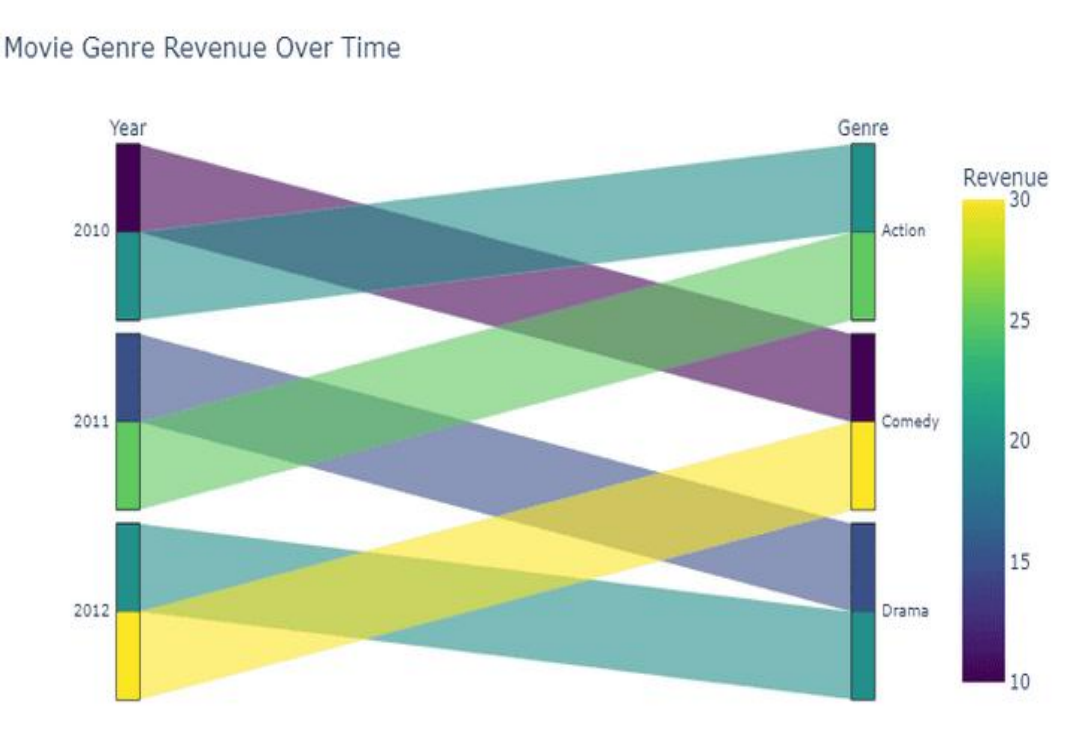
Choosing right chart for data

9. Flow

- Flow charts can be maps that convey movement over time, such as Sankey diagrams. Flow maps include path over time and path between origin and destination charts.

What kind of question does this chart answer?

- What is the longest shipping route?
- How long are people lingering around gates?
- What are the bottlenecks to traffic in the city?



Choosing right chart for data

- Always keep in mind that choosing the right chart is crucial. Think about what your data really wants to say, and then explore Flourish's templates to bring out its potential.

- Flourish helps you transform your data into incredible visualizations and meaningful stories.

3. Best practices for Effective Visualization

Visualizing data is an important aspect of presenting insights clearly.

1. Keep it simple - Data visualization is all about communicating data insights in a way that is both accurate and easy to understand. When creating visuals, less is almost always more. Data overload can quickly lead to confusion, so it's important to only include the most important information.

2. Use intuitive visual cues - The best data visualizations make use of visual cues that are immediately intuitive to the viewer. For example, using different colors to represent different data points can help the viewer more easily understand the relationships between those points.

3. Avoid distracting elements - In addition to keeping the visualization itself simple, it's also important to avoid any extra elements that could serve as distractions. This includes things like excessive use of color, busy background patterns, and cluttered layouts

3. Best practices for Effective Visualization

4. Tell a story - Data visualization is an opportunity to tell a story with the data. An effective visualization will effectively communicate the key insights from the data in a way that is both accurate and easy to understand.

5. Use data points carefully - When using data points in a visualization, it's important to carefully consider which points are most important to include. Data points should be chosen based on their ability to accurately represent the underlying data and tell the story you want to communicate.

6. Use visual hierarchy effectively - An effective data visualization makes use of visual hierarchy to direct the viewer's attention to the most important information. This can be accomplished through the use of things like size, color, and position.

7. Use effective labels - Labels are an important part of any data visualization. They should be clear and concise, and they should accurately describe the data that is being represented.

8. Test data visualizations with users - Before finalizing a data visualization, it's important to test it with actual users. This will help ensure that the visualization is effective in communicating the story you want to tell.

9. Continuously iterate and improve - Data visualization is an iterative process, and it's important to continually strive for improvement. As new data becomes available, or as new insights are gained, be sure to update the visualization accordingly.

4. Data Visualization Libraries

Data visualization provides a good, organized pictorial representation of the data which makes it easier to understand, observe, analyze.

Python provides various libraries that come with different features for visualizing data. All these libraries come with different features and can support various types of graphs.

Matplotlib
Seaborn
Plotly

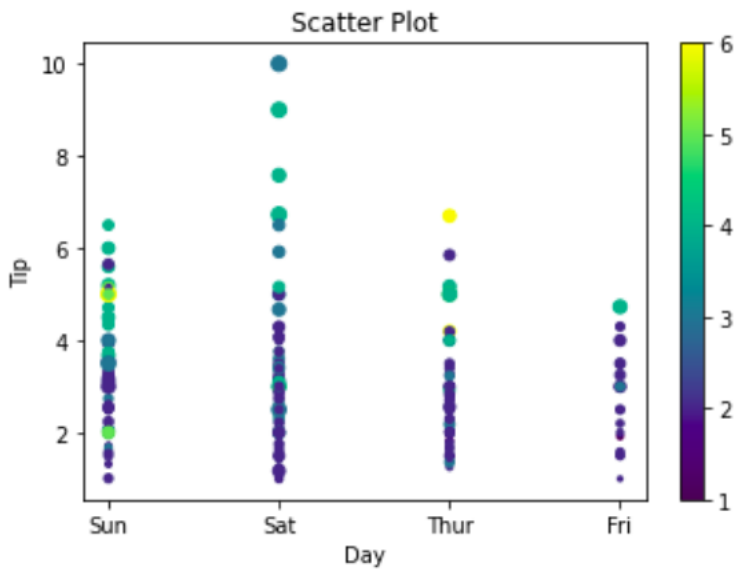
Matplotlib

- Matplotlib is an easy-to-use, low-level data visualization library that is built on NumPy arrays. It consists of various plots like scatter plot, line plot, histogram, etc. Matplotlib provides a lot of flexibility.
- To install this type the below command in the terminal.
- **pip install matplotlib**

- **Scatter Plot**
- Scatter plots are used to observe relationships between variables and uses dots to represent the relationship between them. The **scatter()** method in the matplotlib library is used to draw a scatter plot.

```python
import pandas as pd
import matplotlib.pyplot as plt


# reading the database
data = pd.read_csv("tips.csv")

# Scatter plot with day against tip
plt.scatter(data['day'], data['tip'], c=data['size'],
            s=data['total_bill'])

# Adding Title to the Plot
plt.title("Scatter Plot")

# Setting the X and Y labels
plt.xlabel('Day')
plt.ylabel('Tip')

plt.colorbar()

plt.show()
```

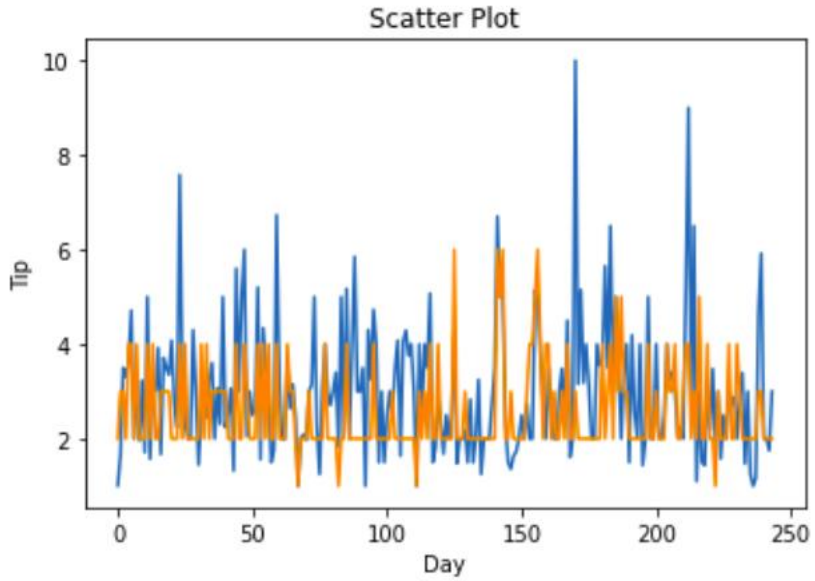
Output:



- **Line Chart**
- Line Chart is used to represent a relationship between two data X and Y on a different axis. It is plotted using the **plot()** function.

```
import pandas as pd
import matplotlib.pyplot as plt


# reading the database
data = pd.read_csv("tips.csv")

# Scatter plot with day against tip
plt.plot(data['tip'])
plt.plot(data['size'])

# Adding Title to the Plot
plt.title("Scatter Plot")

# Setting the X and Y labels
plt.xlabel('Day')
plt.ylabel('Tip')

plt.show()
```

Output:

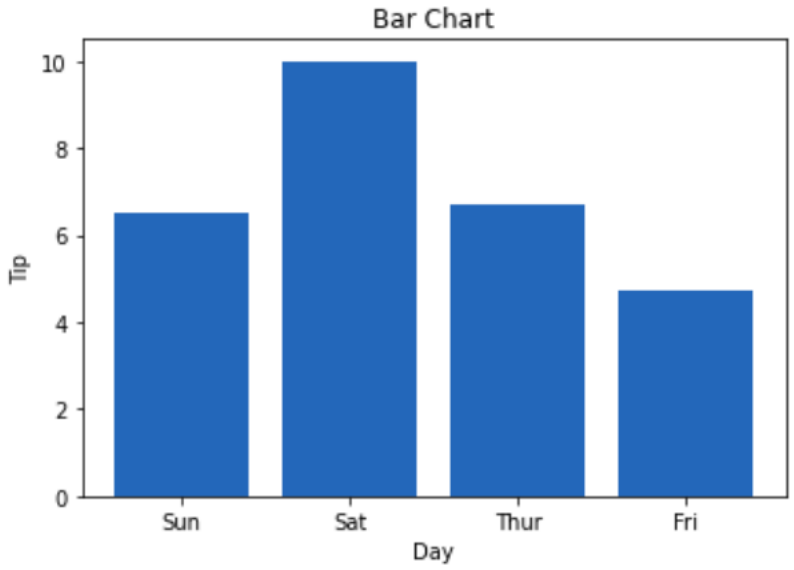


Bar Chart

A bar plot or bar chart is a graph that represents the category of data with rectangular bars with lengths and heights that is proportional to the values which they represent. It can be created using the **bar()** method.

```
import pandas as pd
import matplotlib.pyplot as plt


# reading the database
data = pd.read_csv("tips.csv")

# Bar chart with day against tip
plt.bar(data['day'], data['tip'])

plt.title("Bar Chart")

# Setting the X and Y labels
plt.xlabel('Day')
plt.ylabel('Tip')

# Adding the legends
plt.show()
```

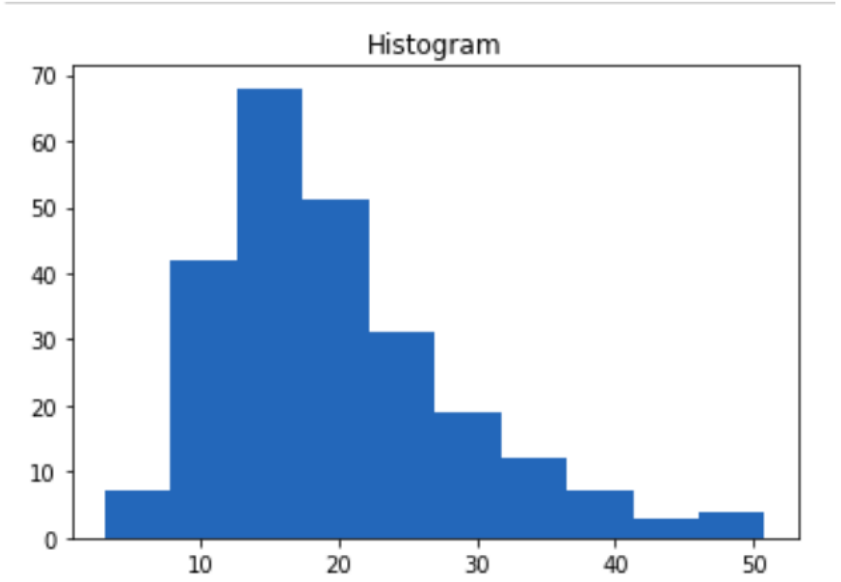
Output:



- **Histogram**
- A histogram is basically used to represent data in the form of some groups. It is a type of bar plot where the X-axis represents the bin ranges while the Y-axis gives information about frequency. The **hist()** function is used to compute and create a histogram. In histogram, if we pass categorical data then it will automatically compute the frequency of that data i.e. how often each value occurred.

```
import pandas as pd
import matplotlib.pyplot as plt


# reading the database
data = pd.read_csv("tips.csv")

# histogram of total_bills
plt.hist(data['total_bill'])

plt.title("Histogram")

# Adding the legends
plt.show()
```

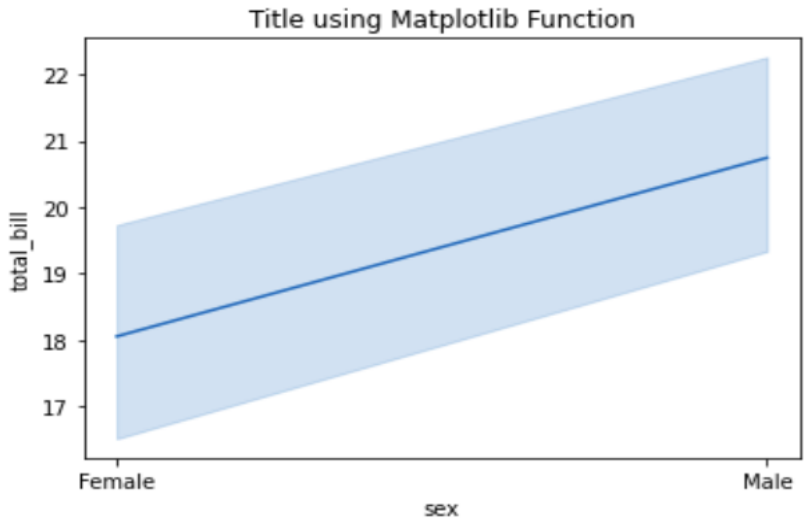
Output:



Seaborn Library for Statistical Analysis

- **Seaborn** is a high-level interface built on top of the Matplotlib. It provides beautiful design styles and color palettes to make more attractive graphs.
- To install seaborn
- pip install seaborn
- Seaborn is built on the top of Matplotlib, therefore it can be used with the Matplotlib as well. Using both Matplotlib and Seaborn together is a very simple process.

Output:

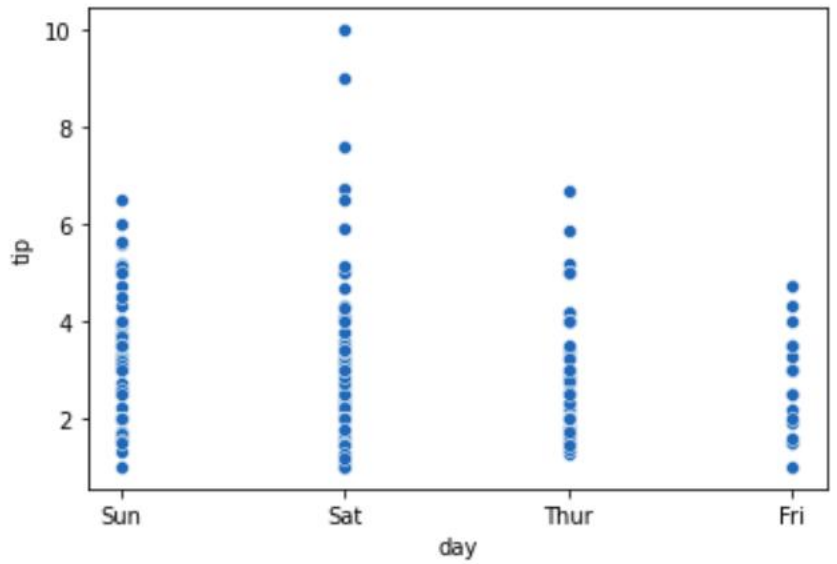



```python
# importing packages
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd


# reading the database
data = pd.read_csv("tips.csv")

# draw lineplot
sns.lineplot(x="sex", y="total_bill", data=data)

# setting the title using Matplotlib
plt.title('Title using Matplotlib Function')

plt.show()
```

- **Scatter Plot**
- Scatter plot is plotted using the **scatterplot()** method. This is similar to Matplotlib, but additional argument data is required.

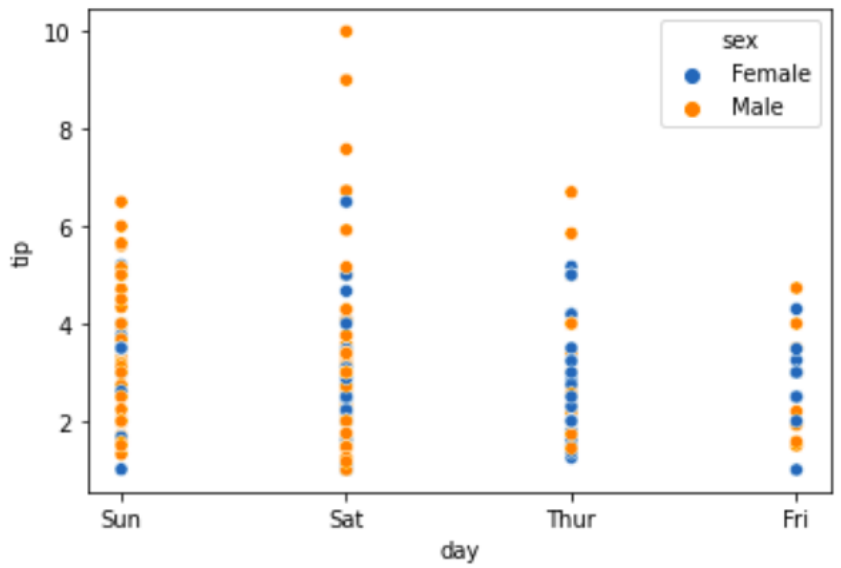
```
# importing packages
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

# reading the database
data = pd.read_csv("tips.csv")

sns.scatterplot(x='day', y='tip', data=data,)
plt.show()
```

Output:



```python
# importing packages
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

# reading the database
data = pd.read_csv("tips.csv")

sns.scatterplot(x='day', y='tip', data=data,
                hue='sex')
plt.show()
```

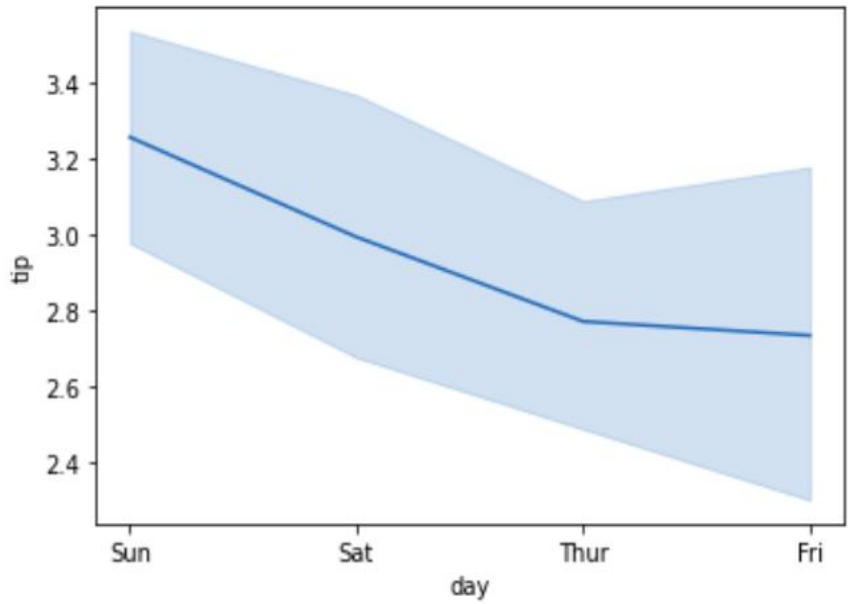
Output:



- **Line Plot**
- Line Plot in Seaborn plotted using the **lineplot()** method.

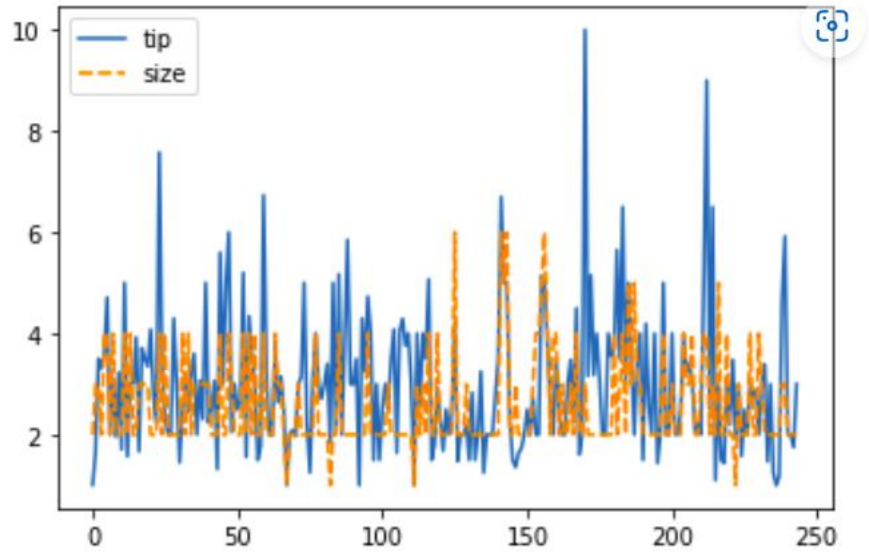
```
# importing packages
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd


# reading the database
data = pd.read_csv("tips.csv")

sns.lineplot(x='day', y='tip', data=data)
plt.show()
```

Output:



```python
# importing packages
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd


# reading the database
data = pd.read_csv("tips.csv")

# using only data attribute
sns.lineplot(data=data.drop(['total_bill'], axis=
plt.show()
```

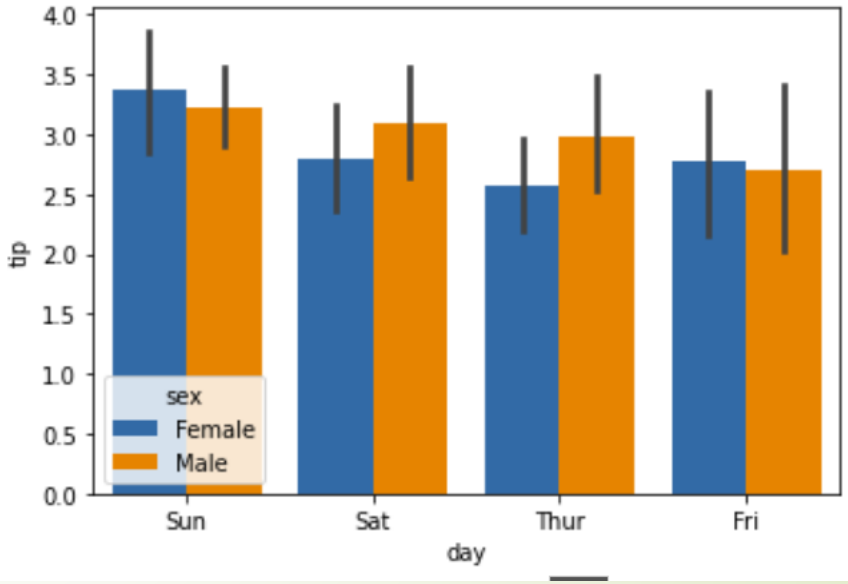
Output:



- **Bar Plot**
- Bar Plot in Seaborn can be created using the **barplot()** method.

```python
# importing packages
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd


# reading the database
data = pd.read_csv("tips.csv")

sns.barplot(x='day',y='tip', data=data,
            hue='sex')

plt.show()
```

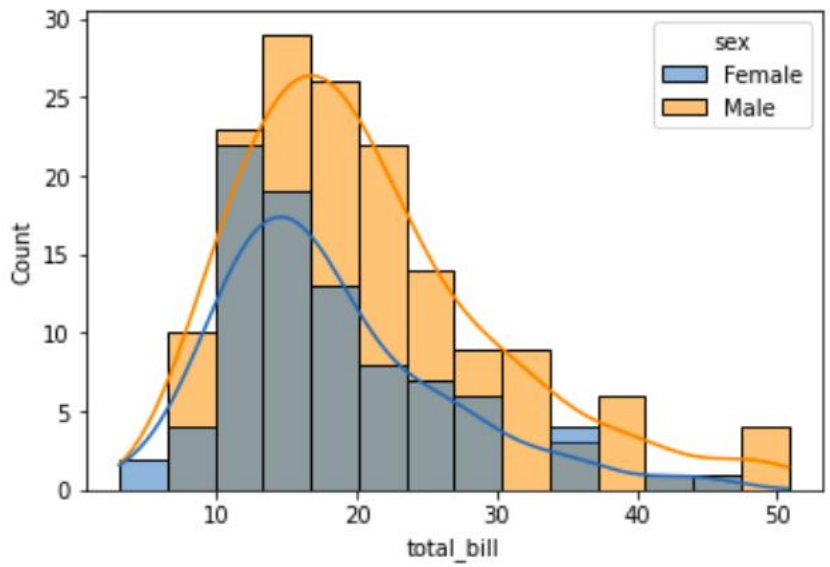
Output:



- **Histogram**
- The histogram in Seaborn can be plotted using the **histplot()** function.

```
# importing packages
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd


# reading the database
data = pd.read_csv("tips.csv")

sns.histplot(x='total_bill', data=data, kde=True, hue='sex')

plt.show()
```

Output:



Plotly Express Library

- **Plotly** library in Python is an open-source library that can be used for data visualization and understanding data simply and easily.
- Plotly has hover tool capabilities that allow us to detect any outliers or anomalies in a large number of data points.
- It is visually attractive and can be accepted by a wide range of audiences.
- Plotly generally allows us endless customization of our graphs and makes our plot more meaningful and understandable for others.
- To Install
- pip install plotly

Package Structure of Plotly

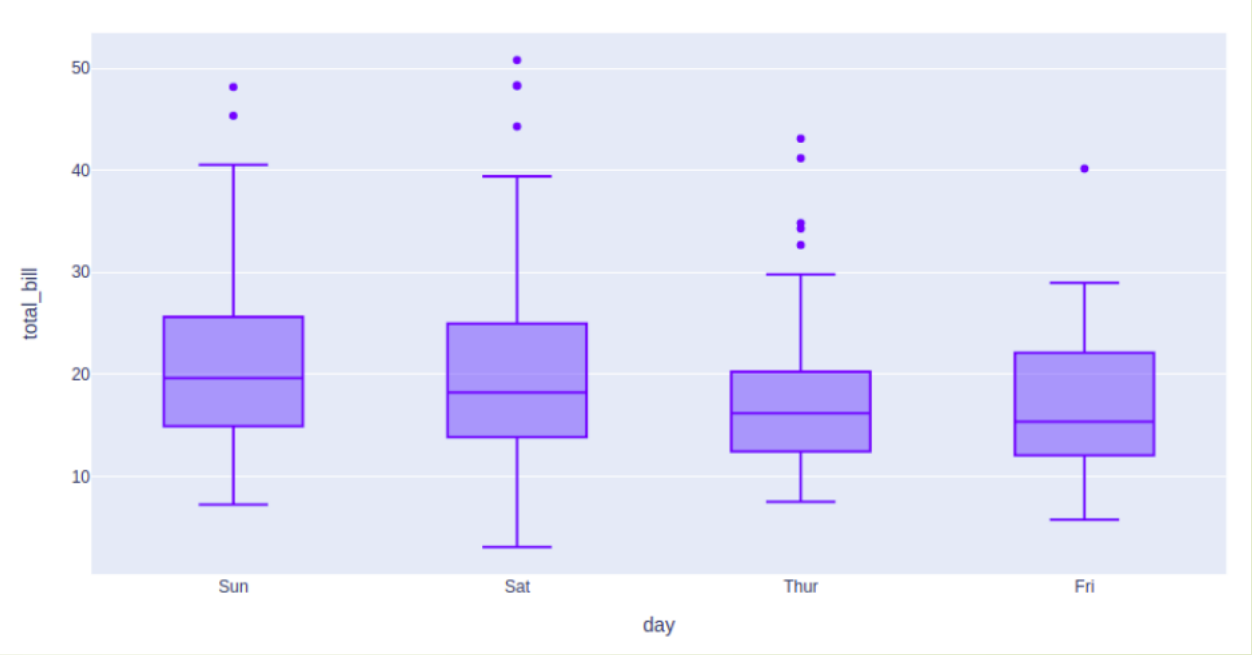
- There are three main modules in Plotly. They are:
- plotly.plotly
- plotly.graph.objects
- plotly.tools
- plotly.plotly acts as the interface between the local machine and Plotly. It contains functions that require a response from Plotly's server.
- plotly.graph_objects module contains the objects (Figure, layout, data, and the definition of the plots like scatter plot, line chart) that are responsible for creating the plots. The Figure can be represented either as dict or instances of plotly.graph_objects.Figure and these are serialized as JSON before it gets passed to plotly.js. Consider the below example for better understanding.

Creating Different Types of Charts

1. Line Chart
2. Bar Chart
3. Histograms
4. Scatter Plot and Bubble charts
5. Pie Charts
6. Box Plots
7. Violin plots
8. Gantt Charts
9. Contour Plots
10. Heatmaps
11. Error Bars
12. 3D Line Plots
13. 3D Scatter Plot Plotly
14. 3D Surface Plots

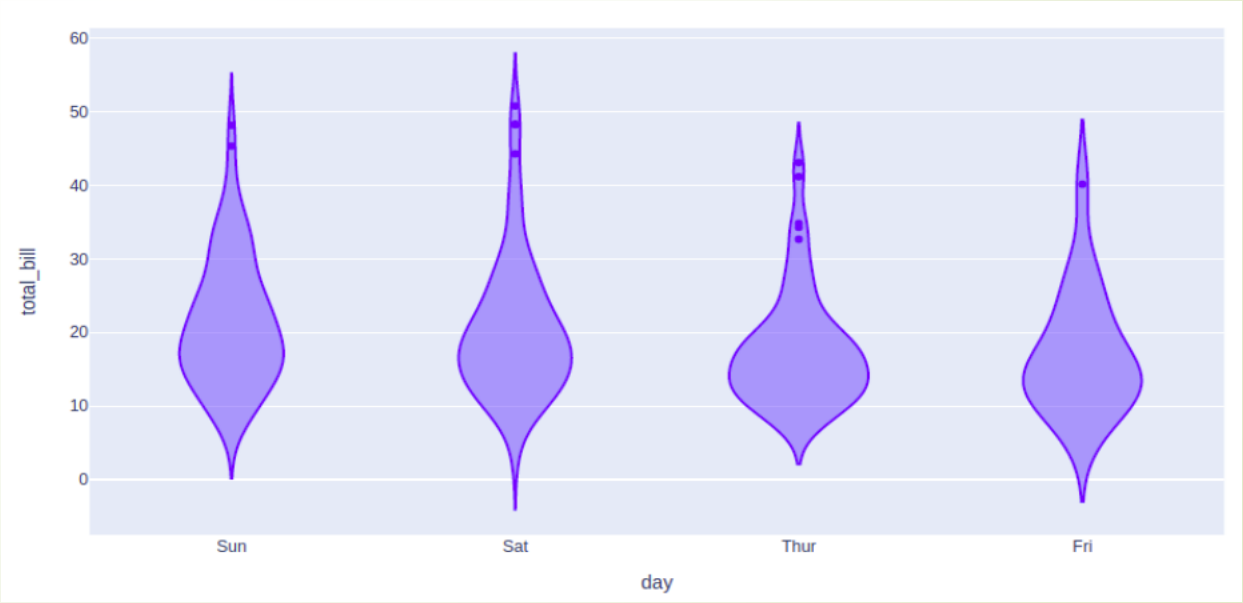
Box Plots

- A **Box Plot** is also known as Whisker plot is created to display the summary of the set of data values having properties like minimum, first quartile, median, third quartile and maximum.
- In the box plot, a box is created from the first quartile to the third quartile, a vertical line is also there which goes through the box at the median. Here x-axis denotes the data to be plotted while the y-axis shows the frequency distribution.



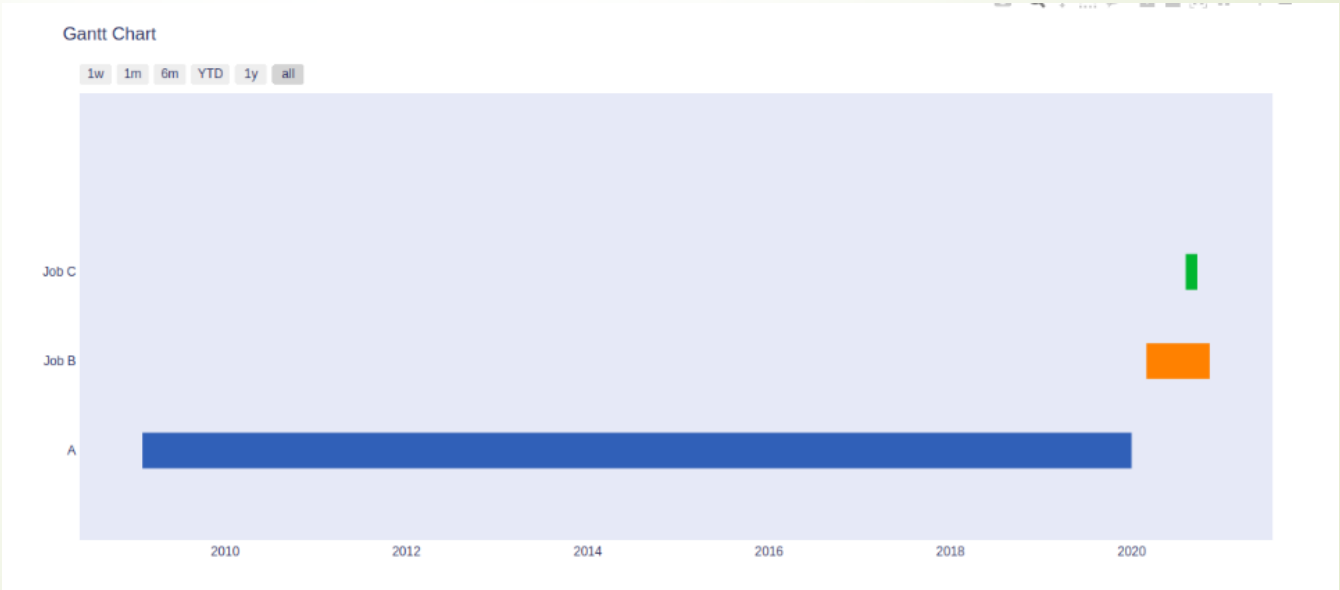
Violin plots

- **A Violin Plot** is a method to visualize the distribution of numerical data of different variables. It is similar to Box Plot but with a rotated plot on each side, giving more information about the density estimate on the y-axis. The density is mirrored and flipped over and the resulting shape is filled in, creating an image resembling a violin.
- The advantage of a violin plot is that it can show nuances in the distribution that aren't perceptible in a boxplot. On the other hand, the boxplot more clearly shows the outliers in the data.



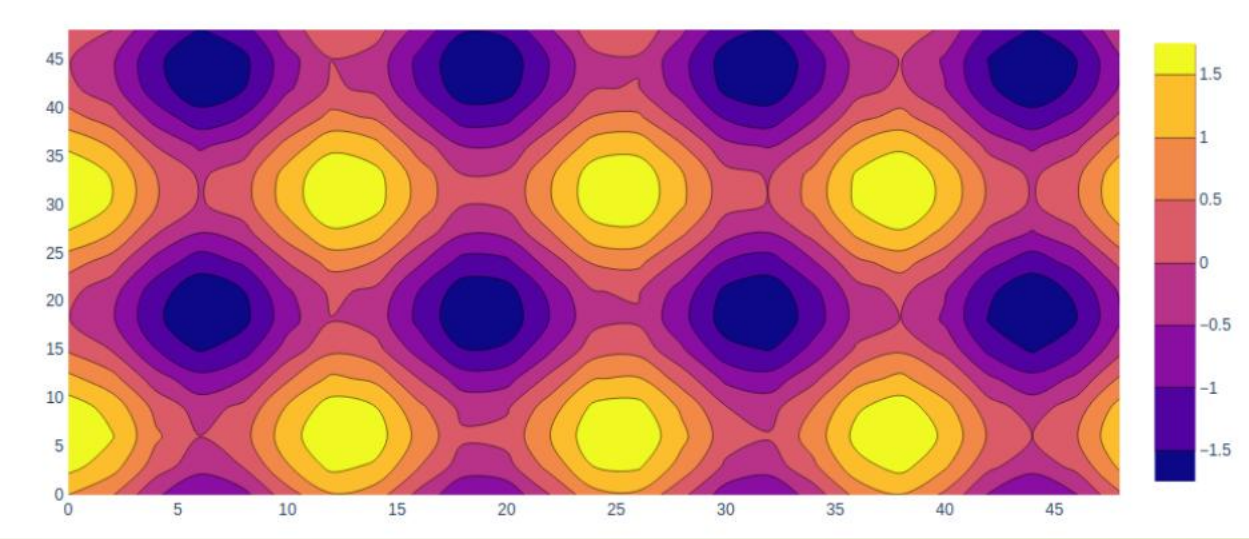
Gantt Charts

- **Generalized Activity Normalization Time Table (GANTT) chart** is type of chart in which series of horizontal lines are present that show the amount of work done or production completed in given period of time in relation to amount planned for those projects.



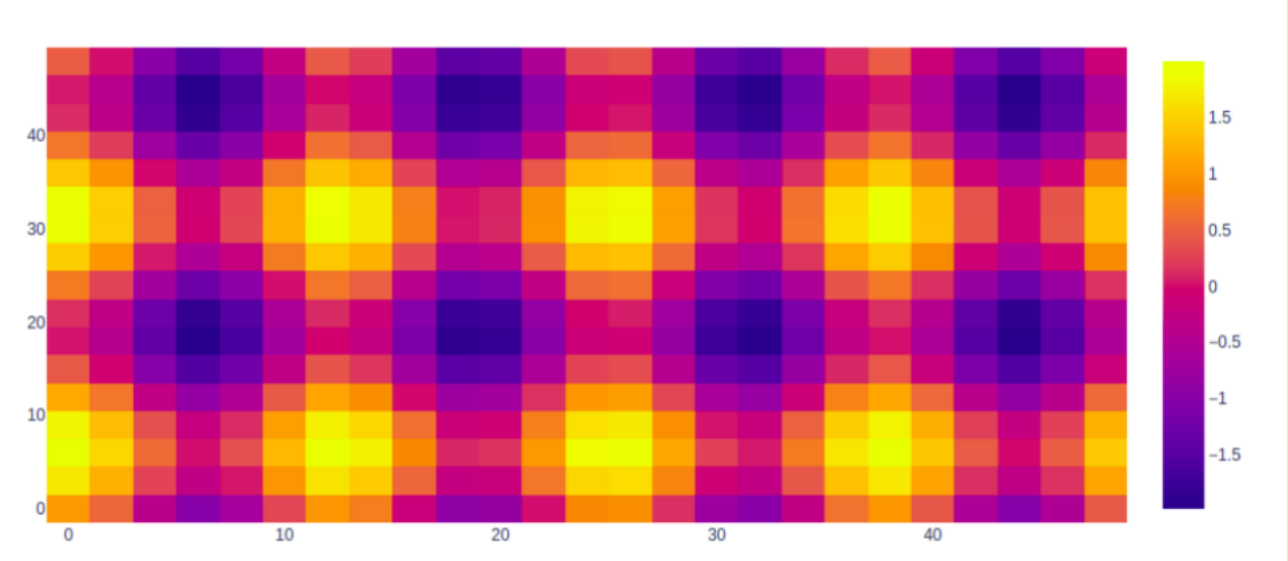
Contour Plots

- A **Contour plots** also called level plots are a tool for doing multivariate analysis and visualizing 3-D plots in 2-D space. If we consider X and Y as our variables we want to plot then the response Z will be plotted as slices on the X-Y plane due to which contours are sometimes referred as Z-slices or iso-response.



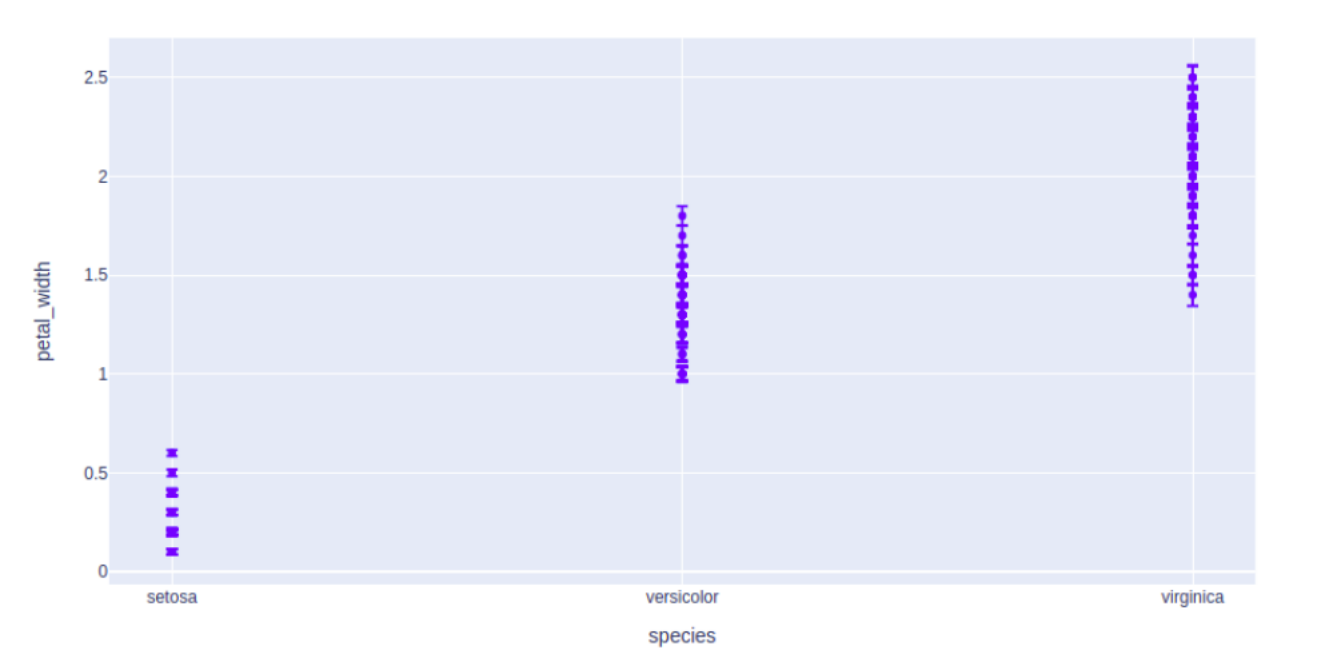
Heatmap

Heatmap is defined as a graphical representation of data using colors to visualize the value of the matrix. In this, to represent more common values or higher activities brighter colors basically reddish colors are used and to represent less common or activity values, darker colors are preferred. Heatmap is also defined by the name of the shading matrix.



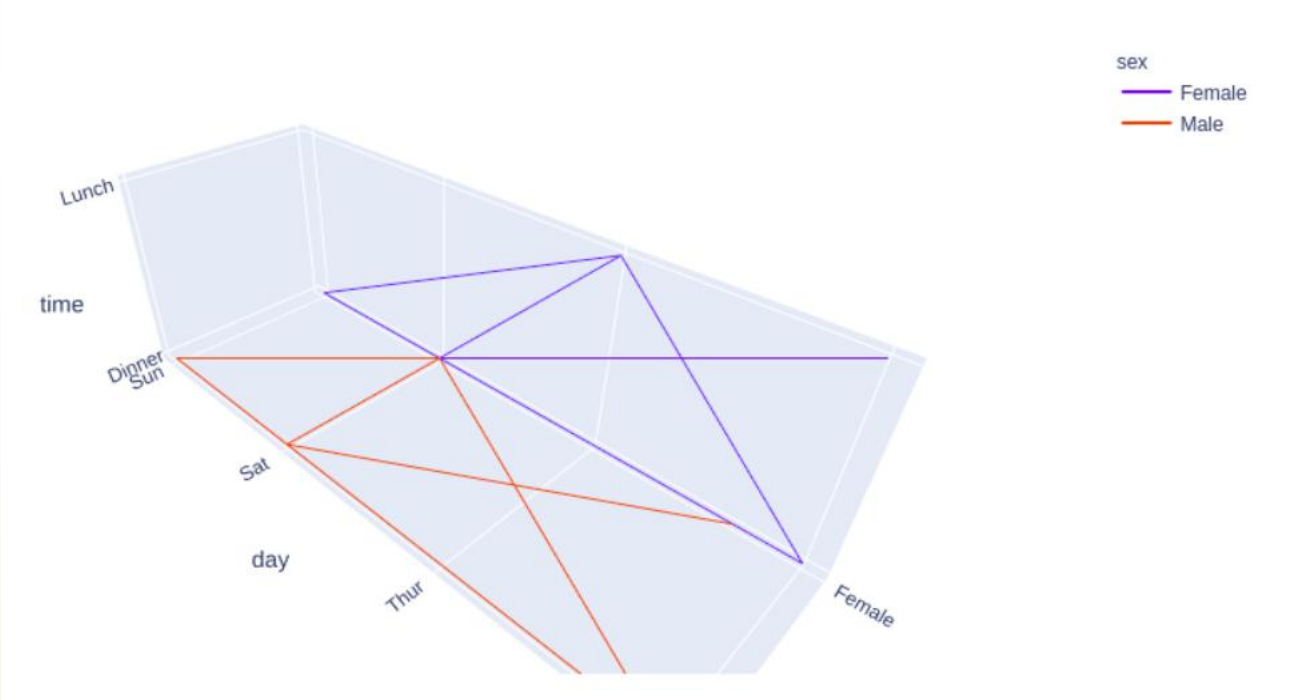
Error Bars

- For functions representing 2D data points such as px.scatter, px.line, px.bar, etc., **error bars** are given as a column name which is the value of the error_x (for the error on x position) and error_y (for the error on y position). Error bars are the graphical presentation alternation of data and used on graphs to imply the error or uncertainty in a reported capacity.



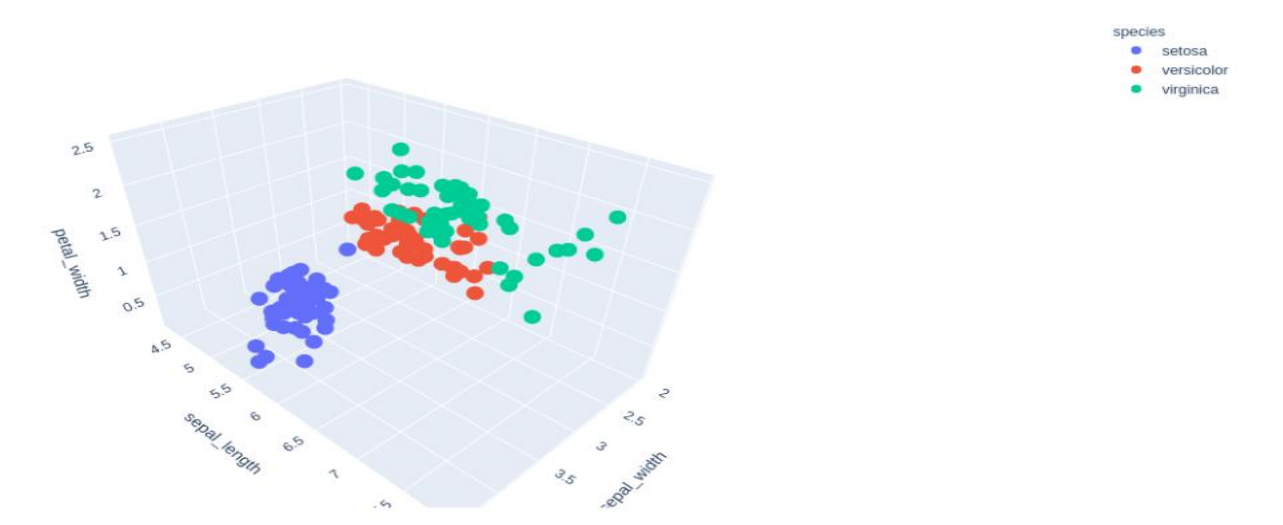
3D Line Plots

- Line plot in plotly is much accessible and illustrious annexation to plotly which manage a variety of types of data and assemble easy-to-style statistic. With **px.line_3d** each data position is represented as a vertex (which location is given by the x, y and z columns) of a polyline mark in 3D space.



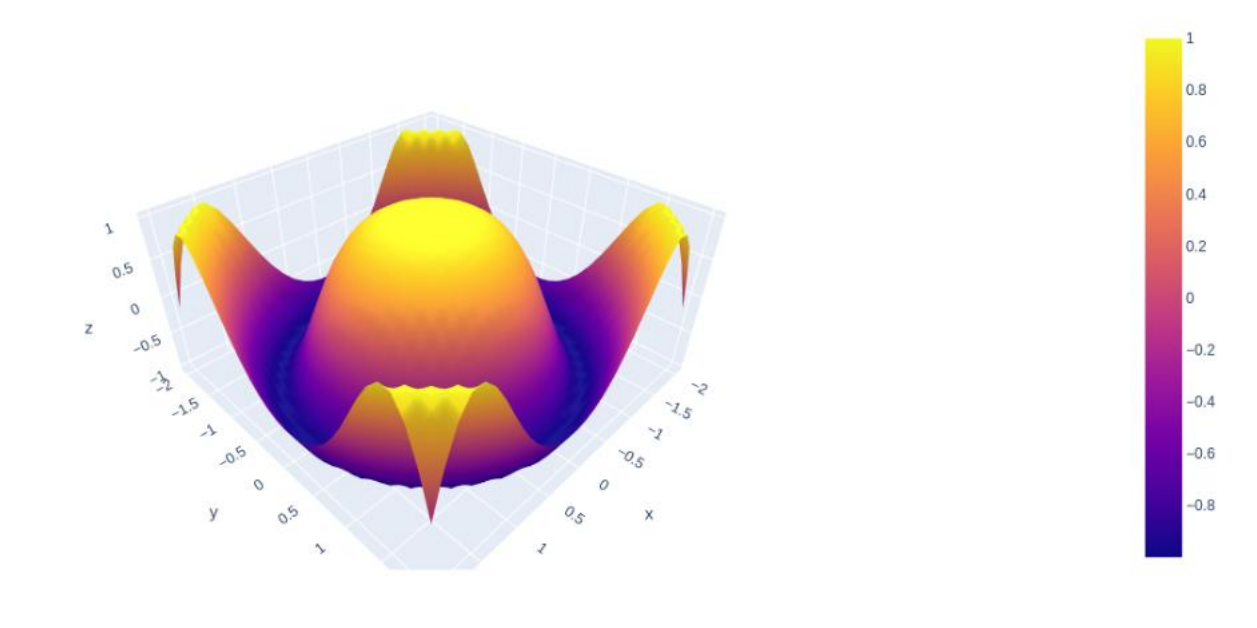
3D Scatter Plot Plotly

- **3D Scatter Plot** can plot two-dimensional graphics that can be enhanced by mapping up to three additional variables while using the semantics of hue, size, and style parameters. All the parameter control visual semantic which are used to identify the different subsets. Using redundant semantics can be helpful for making graphics more accessible. It can be created using the scatter_3d function of plotly.express class.



3D Surface Plots

- **Surface plot** is those plot which has three-dimensions data which is X, Y, and Z. Rather than showing individual data points, the surface plot has a functional relationship between dependent variable Y and have two independent variables X and Z. This plot is used to distinguish between dependent and independent variables.



Interacting with the Plots

- Plotly provides various tools for interacting with the plots such as adding dropdowns, buttons, sliders, etc. These can be created using the update menu attribute of the plot layout.
- **Creating Dropdown Menu in Plotly**
- A **drop-down menu** is a part of the menu-button which is displayed on a screen all the time. Every menu button is associated with a Menu widget that can display the choices for that menu button when clicked on it. In plotly, there are 4 possible methods to modify the charts by using update menu method.

- **restyle:** modify data or data attributes
- **relayout:** modify layout attributes
- **update:** modify data and layout attributes
- **animate:** start or pause an animation

Adding Buttons to the Plot

- In plotly, **actions custom Buttons** are used to quickly make actions directly from a record. Custom Buttons can be added to page layouts in CRM, Marketing, and Custom Apps. There are also 4 possible methods that can be applied in custom buttons:

- **restyle:** modify data or data attributes
- **relayout:** modify layout attributes
- **update:** modify data and layout attributes
- **animate:** start or pause an animation

Creating Sliders and Selectors to the Plot

- In plotly, the **range slider** is a custom range-type input control. It allows selecting a value or a range of values between a specified minimum and maximum range. And the range selector is a tool for selecting ranges to display within the chart. It provides buttons to select pre-configured ranges in the chart. It also provides input boxes where the minimum and maximum dates can be manually input.